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An unsharp masking filter technique with weighted bilinear interpolation for endoscopy image quality and sharpness enhancement

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Abstract
The traditional Endoscopy method is a golden standard technique for observing gastrointestinal tract diseases. To properly identify gastrointestinal diseases, doctors mostly need good-quality images. The quality of the endoscopic images is mostly low in different hospitals due to the limited camera performance, complex gastrointestinal (GI) tract environment, and poor illumination. Most of the existing techniques often use the information of the image or multiple images of the same scene to offer enhancement. To overcome the issues associated with low contrast and noisy endoscopic images, this paper presents a novel image enhancement technique mainly developed to improve diagnostic accuracy and GI tract visualization. The proposed model begins by converting the images to the Hue Saturation Value (HSV) space to independently process the intensity component. In this proposed study, we applied a half-unit Weighted Bilinear Interpolation (Hu-WBI) algorithm method combined with the Contrast Limited Adaptive Histogram Equalization (CLAHE) on the inverted image intensity component by optimizing contrast while controlling noise. To achieve seamless contrast enhancement, the Hu-WBI is applied to overcome the over-enhanced images with artifact issues associated with CLAHE. To enhance the sharpness of the final images, an unsharp mask filter integrated with a Super Resolution Generative Adversarial Network (SRGAN) is applied. The hybrid methodoly preserves a details of high-frequency and it reduces excessive noise amplification Since the normalized intensity component is removed from the Low Pass filter, the final enhanced GI images are clearer and more precise which is widely adopted clinical technique for reading in the Computer Aided Diagnostic system. The proposed model is deployed in the Internet of Things (IoT) based cloud platform to improve the diagnostic capabilities of the medical images and demonstrate its practical impact on telemedicine. The proposed method achieves a very low error rate under extensive simulation experiments. The simulation results were conducted on two different datasets utilizing different performance evaluation metrics such as different metrics such as Structural Similarity Index (SSIM), Mean Squared Error (MSE), Adjusting image Intensity values (ADJ) and Feature Similarity Index (FSIM). When compared to the existing techniques, the proposed model improves the FSIM and IRMLE values by up to 32%. The proposed model offers a PSNR, SSIM, and LPIPS value of 20.1212, 0.8945, and 0.1456 when evaluated using the CVC clinic dataset.

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Keywords CLAHE · Weighted bilinear-interpolation · Unsharp masking filter · Inverse intensity · Super resolution generative adversarial network

1 Introduction

In medical diagnostics, processing images is crucial. Gastrointestinal (GI) track-related infections and diseases are becoming common threats to life in recent days [1]. Medical procedures have made extensive use of both traditional endoscopy and Wireless Capsule Endoscopy (WCE) techniques. These images help physicians in the observation of intestinal tract tissues, diseases, and some abnormalities present in the tissues of the abdomen. Early diagnosis of GI tract infections or diseases via endoscopy or WCE is widely adopted clinical technique for extended life [2]. Endoscopy is a method that involves sending a tiny lens camera through the stomach and capturing images via computer systems [3].

A Digital computerized method is needed to get information to the fullest from these images. Even though endoscopy images contain information on tissue samples, images are subject to noise, low contrast, and some other problems. The images are blurred due to noise and repositioning of the camera lens, over-exposure of white light, and missing dark areas during camera movement [4, 5]. As a result, noise in the image significantly degrades its quality by causing content blur and features to be lost in dark areas. Shah et al. [6] addressed the issue of noise removal from the enhanced image using wiener filter, and linear contrast methods. Image contrast enhancement and improvement is the only solution to obtain detailed information and clearance relating to the vessels and structures of the gastrointestinal tract [7]. As a result, improving endoscopy and WCE imagery is crucial for the diagnosis and classification of infections and illnesses.

Numerous image-enhancement (IE) methods are being put forth to raise the standard of endoscopic images. Histogram Equalization (HE), Adaptive Histogram Equalisation (AHE), Gaussian Filters, and Contrast Limited Adaptive Histogram Equalisation are the most often utilized IE approaches in medical image detection [8, 9]. Several HE-based techniques utilize linear and/or bilinear interpolation (BIL) approaches to achieve better results than other HE-based techniques [9]. Bilinear interpolation with cubic interpolation can be used to deal with the artifacts generated by HE-based methods while converting the image from an RGB color channel to another color space. For instance, AHE uses linear and bilinear interpolation methods which divide the entire image into small distinct blocks and apply it in each of the sections. A revolutionary approach for brightness enhancement and modifications that maintains the dynamic histogram equalization and lighting is called Brightness Preserving Dynamic Fuzzy Histogram Equalization (BPDFHE) [10]. Despite this algorithm being very good at preserving the details of the image and its intensity, it fails in some cases. Details are provided in the experiments and discussions section.

An enhanced form of AHE called contrast-limited Adaptive Histogram Equalisation (CLAHE) [11] was created to reduce noise in images and provide precise, detailed data. Bilinear interpolation in HE is used in AHE techniques, which do not entirely suppress noise. This paper presents a novel approach that addresses multiple issues with image quality improvement in endoscopic images by integrating contrast enhancement, noise reduction, and sharpening modules. Initially, images are collected from the clinical (S.K Crystal Clinic, Chennai) and online endoscopy image datasets. Conventional models directly improve RGB images and it often leads to improper contrast handling and color distortion. The proposed model converts the image into HSV color space to separate the intensity component from the color information. In this way, we can control the areas that need brightness and contrast improvement without altering the color distribution. The CLAHE model is only applied after the intensity component is normalized and inverted to offer better contrast without increasing the noise.

The contrast-limited Adaptive Histogram Equalization (CLAHE) is integrated with bilinear interpolation to improve the image contrast and preserve the diagnostic information. The Hu-WBI is used to eliminate the issues associated with the CLAHE technique such as over-enhanced images with artifacts. The Hu-WBI strategy doubles the image size and recomputes the pixel values resulting in seamless contrast enhancement. The traditional High Pass Filters (HPF) and unsharp masking filters are highly noise-sensitive. Hence, instead of applying HPF directly to the images, we are passing the enhanced image through SRGAN. To sharpen the images and convert them into high resolution, a Super Resolution Generative Adversarial Network (SRGAN) is integrated with an unsharp mask filter resulting in improved visual results for diagnosis.

The proposed model is also deployed in an Internet of Things (IoT) platform which also improves clinical decision-making, and patient care, and provides real-time insights via high-quality endoscopic images. Unlike traditional methods, the proposed model offers remote access and real-time processing. The proposed model is highly suitable for telemedicine applications where the endoscopic images can be remotely enhanced with limited time. Finally, we can achieve high-quality enhanced color images with the help of improved intensity. This method's low computing cost and ease of use enable additional extensions for real-time applications. This method can be used as a computer-aided diagnosis tool which will give enhanced noise-free smoothening and sharpened images. The main contributions of this paper are presented as follows:

- This paper presents a novel hybrid algorithm for improving the image quality of endoscopic images by incorporating contrast enhancement, noise reduction, and sharpening modules.
- A CLAHE with Bilinear Interpolation is presented for contrast enhancement which preserves the crucial information even while improving the contrast.
- The super Resolution Generative Adversarial Network (SRGAN)is integrated with the unsharp mask filter to improve image sharpness.
- The practical impact of the proposed model is validated by implementing it in an Internet of Things (IoT) based cloud computing network to improve clinical decision-making and patient care.

Novelty Introduce Hu-WBI differs from traditional bilinear and threshold interpolation methods by computing intermediate pixel values with half unit weighting, which is effectively minimized over-enhancement artifacts that introduces CLAHE. The proposed approach adjusts the intensity based on threhsolding while maintaining structural fidelity in endoscopic images and diagnostic visibility.

This paper is organized in the following order. Related works on image processing and enhancement methods are covered in Sect. 2. The suggested methodology and algorithm are shown in Sect. 3. Discussions and experiments are included in Sect. 4. The conclusions are presented in Sect. 5 along with the direction of future research.

2 Related works

Geleijnse et al. [12] suggested an Edge Enhancement Optimization (EEO) model for improving the contrast of endoscopic images. They included three numerical elastic endoscopic models. To evaluate the effects of edge enhancements they conducted two types of experiments: in vitro and in vivo. This study is mainly conducted to identify the impact of edge enhancement, sharpness, and noise. Using 39, ENT professionals the vivo images were evaluated and compared using a pairwise approach. They observed that an increase in the edge enhancement levels within a range of 0.8–1.2 resulted in an improvement in the noise and sharpness of the image. The specialists identified the optimal edge enhancement level to be within 0.7–0.9. The demerits of this work are the risk of over-enhancement, loss of subtle gradations, and reduced naturalness.

Moghtaderi et al. [13] presented a guided filter decomposition and wavelet transform (GFD-WT) for endoscopic image enhancement. Endoscopic treatment efficiency, medical decision-making, and patient care have significant effects based on the images' visual quality. The image fusion with the enhancement of endoscopic images was the approach suggested. The global and local contrasts were complementary to one another. The endoscopic images' visual quality is enhanced via this method. The GFD-WT was subjected to each sublayer using appropriate fusion rules to result in a better-quality image. To confirm the efficacy of GFD-WT, endoscopic images that tracked the upper gastrointestinal tract were analyzed. This work offered advantages like enhanced structural details, image quality preservation, and improved contrast, but, it faced challenges in terms of dependency on parameter selection, complexity, and computational demand.

Based on the brightness classification, Nie et al. [14] presented a technique to eliminate specular reflections from endoscopic images. This approach mainly focused on improving the high-resolution texture structure information and the operating efficiency. The image brightness adjustment and adaptive threshold function were included to improve the brightness component and classification. The challenges were information loss risk, computational overhead, and compatibility issues. Image enhancement techniques can be roughly categorized into two domains: the spatial domain and the frequency domain [15]. Under the Spatial domain, direct image pixel transformation is done to get the desired enhancement. The image is initially moved into the frequency domain, where the Fourier transform is calculated for the image. The enhanced image is then obtained by multiplying the image result by filters and performing an inverse Fourier transform on the image. Histogram Equalization (HE) methods are one of the basic fundamental image enhancement methods for images with simplicity, but they enhance noise too [16]. The attributes of the color channels (R, G, and B) are changed when histogram equalization is applied to RGB images, which results in modifications to the image's hues. It's better to convert images from RGB to LAB color space while working with HE techniques because HE will change the properties of the RGB color channel.

Zhang et al. [17] have presented a study that uses an enhanced weighted guided filter algorithm with unsharp masking to improve the outlines and features of vasculature in endoscopic pictures. The mentioned algorithm enhances edges with noise reduction and minimizes halos to give the proper structure of endoscopic images. Tomoya Sato et al. [18] addressed the issues of endoscopy image processing such as non-uniform illumination by proposing texture and color enhancement imaging techniques. In this study, color augmentation and retinex-based enhancement are coupled to emphasize tissue color variations. CLAHE and AHE are another method that works on limiting contrast and enhancing the image intensity. It provides a histogram clipping limit for suppression of the noise problems.

The full image will be split into smaller sections of tiles since CLAHE works on a small portion of the image known as tiles. This enables CLAHE to precisely regulate the contrast and noise enhancement across the image. The Threshold Weighted-bilinear algorithm(TWB) is an extended approach of half unit weighted algorithm which helps preserve Hue's by avoiding overexposure and enlargement of specular light spots [19]. This algorithm works by increasing the intensity content by decreasing its average weights. This algorithm works well in enhancing darker areas but it's Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE) score is less. The Unsharp Masking(UM) Filter algorithm method processes the high and low frequencies components of the image sharply which gives detailed enhancement with rich information about the image. However, these algorithms generate artifacts and give artificial color to the enhanced images. Retinex-based techniques, such as Robust Retinex [20], make a significant enhancement of images and improvement in human visualization. However, they typically produce halo effects.

Additionally, these algorithms are highly time-consuming and complicated. In short, the above-mentioned techniques and methods focus on brightness enhancement, intensity, and illuminance correction and improvements. The main drawback of the mentioned methods lies in the noise getting amplitude and blurred effects due to disturbed camera sensors for the real-time datasets. It is been clear that the choice of the algorithm that is taken into consideration depends entirely on endoscopy image specifications. The taxonomy of the existing techniques is listed in Table 1.

3 Proposed framework

The proposed overall workflow of the image enhancement framework for SRGAN based sharpening is depicted in Fig. 1. The proposed work incorporated into various stages, (i) convert the input RGB image into HSV color space and Value (V) channel is extracted for processing, (ii) Based on V-channel, CLAHE performs contrast enhancement, (iii) apply Hu-WBI along with unsharp masking to suppress artefacts and enhance edge sharpness and (v) perform SRGAN-based super resolution followed by the reconstruction of of HSV-RGB thereby producing an improved output images. The workflow of the proposed method goes in four phases, including:

- Data Collection and preparation.
- Convert the RGB image into HSV color space and enhancement is applied to the Value (V) component.
- Applying CLAHE to the inversed Intensity value with Half-unit Weighted Bilinear Interpolation (WBI).
- Applying Unsharp Masking Filter.

Table 1 Summary of existing literature

Reference	Technique used	Purpose	Advantages	Limitations
[12]	EEO	To enhance the clarity of edges in endoscopic images	Improvement in edge enhancement	Loss of subtle gradations
[13]	GFD-WT	To improve the visual quality of endoscopic images	Improved contrast and image quality preservation	Dependence on parameter selection
[14]	Specular reflection removal	Restores high-resolution texture	Improves brightness and eliminates specular reflections	Information loss
[15]	HE	Enhancing the image in spatial and frequency domains	Simplified enhancement process	Modification in image hues
[17]	The enhanced Weighted Guided Filter (WGF) algorithm	To enhance the outline and features of endoscopic images	Reduced halos and noise reduction	Results in artifacts
[18]	Retinex-based enhancement	To improve the texture and color of endoscopic images	Overcomes the non-uniform illumination problem	Time-consuming
[19]	TWB	Improves image intensity and contrast	Accurate noise control and contrast	Complex process (Splitting images into tiles)
[20]	Robust retinex methodology	Utilizes an additional noise mask when compared to the traditional retinex algorithm	Can be used in diverse image enhancement applications	Amplitude and blur effects

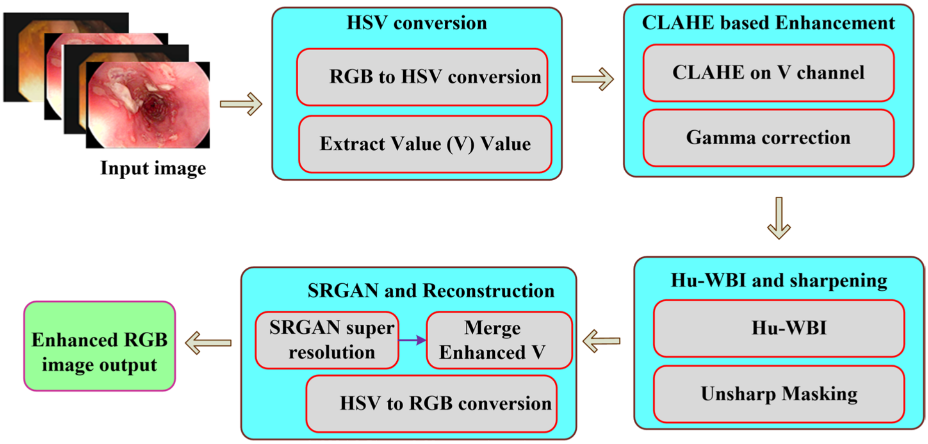

Fig. 1 Overall workflow of the proposed image enhancement framework for SRGAN based sharpening

The first phase includes data collection and data cleaning which is the major and foremost task to be done. In the second stage, the input image is converted to HSV (Hue, Saturation, Value) and the enhanced process is applied to the Value channel that represents brightness, where Hue represents the pure color (0–360°), saturation represents color intensity (0–100%) and Value representsbrightness (0–100%). Hue and saturation deal with the color component of an image while intensity deals with the brightness of the image. The contrast improvement process is made easier by the conversion of RGB color space to HSV color space. After the Value component is retrieved, the computation of the normalization and inversion of the normalized intensity component is done. Then CLAHE with Hu-WBI is applied to the achieved result in the third phase of work. In this phase, the image size is doubled and the low contrast value of the pixel is calculated through WBI. In the fourth phase of work, the output image of the third phase is given as input to the High Pass Filter (HPF) named Unsharp masking Filter and SRGAN which extracts better information from high frequency components (masking unit). An unsharp mask filter is used for sharpening the image and providing the image for machine readability in a better way.

3.1 Data collection and preparation

In this work, endoscopy image datasets were collected from S.K Crystal Scans Clinic Hospital, Chennai they were 360 in count with different dimensionalities, publicly available datasets that are used for evaluation of the proposed method. The online datasets are CVC-Clinic DB [21], ETIS-Larib [22], and kvasir-dataset [23]. All images are in RGB format. A summary of the datasets collected is shown in Table 2. The images are reshaped into 512 × 512. Some of the reference sample images are shown in Fig. 2 (a)-(d).

3.2 Converting RGB image to HSV colour space component

Luminance or intensity values could not be separated from the color space and therefore it's very difficult to get the information if the brightness value is too high. When the input image is converted to HSV (Hue- Saturation- Value), the hue, saturation, and Intensity val-

Table 2 Datasets with their count and type of class

Type of class	Dataset name	Number of Images
Esophagitis	Kvaiser-dataset	450
Dyed-lifted-polyps	Kvaiser-dataset	450
Polyps	CVC-Clinic DB	613
Helicobacter Pylori	Real-time dataset	82

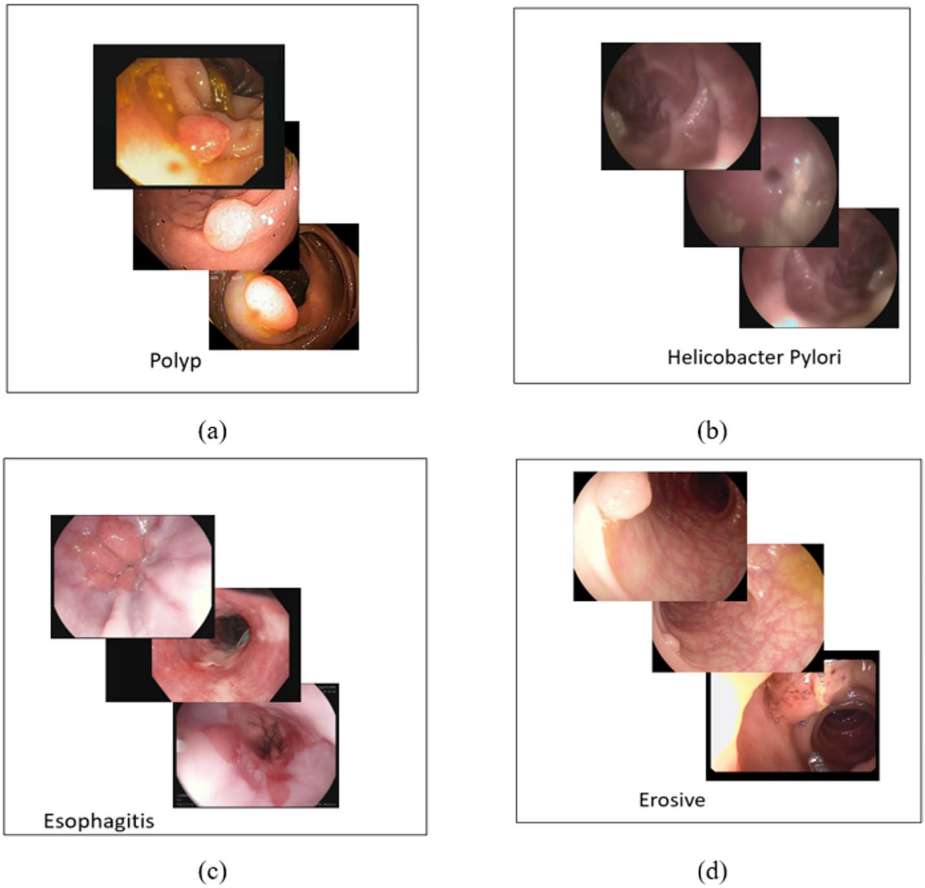

Fig. 2 Sample dataset images

ues could be separated which were independent. Separated Value provides a low impact on brightness changes. Hue and saturation values remain unaltered. The components of hue, saturation, and value are denoted, respectively, as H(a, b), S(a, b), and I(a, b). The intensity component $I_I(a,b)$ was normalized between 0 and 1. Hence the pixel values range between 0 and 1. The normalization of the Intensity value can be calculated by Eq. (1).

$$I_I(a,b) = \frac{I(a,b)}{I_I\,(Max)} \tag{1}$$

Where, $I_I(a,b)$ is the normalized Value, $I_I\,(Max)$ is the maximum Value component (Value component value ranges from 0 to 255). After normalization, the value will range between 0 and 1. The inverse of the normalized Value component was taken to remove the noise from the intensity-enhanced image. The Inverse of the normalized intensity component is given in Eq. (2).

$$I' = 1 - I_I(a,b) \tag{2}$$

Gamma correction was performed in the intensity component to improve the contrast. Larger (when $\gamma > 1$) and smaller (when $\gamma < 1$) values of intensity components were enhanced based on the value of gamma which is denoted as γ. Gamma defines the shape of the curve that depicts how the input and output images are related. It is mathematically represented as

$$Y = C * M^{\gamma} \tag{3}$$

Where C is a constant, Y is the enhanced output image, M is the input image and γ is the gamma value. The inverse intensity image is given as input to the CLAHE method, which is the next method in the process.

3.3 Contrast limited adaptive histogram equalization with WBI

CLAHE a method seen in the histogram equalization provided a clipping limit for the histogram for a reduction in noise [11]. It applies HE all over the image by dividing the image into small tiles. CLAHE was applied with Hu-WBI to the inverse normalized image [24]. The size of the image was doubled and Hu-WBI was applied to the adjacent pixels for the extraction of the information from the hidden darkened image pixels. Equation 4 provides a numerical explanation for the Hu-WBI algorithm:

$$H(a,b) = \frac{\sum_{t=1}^{4} P_n}{2} \tag{4}$$

Where, P_n represents the adjacent four pixels' locations.

$H(a,b)$ represents the intermediate location between the pixel in the upgraded pixel image (Fig. 3).

Table 3, provides a pictorial view of input matrix (i) which is a (3×3) image pixel, (ii) shows the upscaled image matrix through the Bilinear interpolation operation. (iii) shows enhanced image pixels of the upscaled image through Hu-WBI operation.

Values for the unknown pixels for an upscaled image were computed using bilinear interpolation. Consider, Pi is the unknown pixel, Ai, Bi, Ci, and Di is the nearest known pixel, then the weighted average of the known nearest pixels were calculated in both directions of the new point for example, in Table 3, the new pixel "7" was calculated based on calculating the distance from pixel value "12' to "2" that is [½(12) + ½(2)].

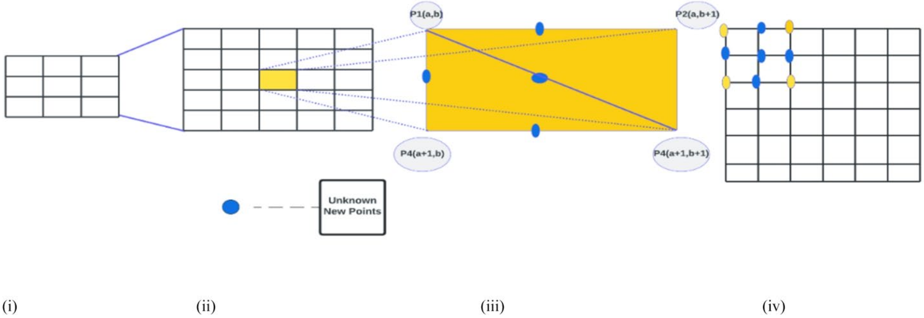

Fig. 3 (**i**) Input Image sample of pixel representation (**ii**) Upscaling of 3 × 3 pixel by 5 × 5 (**iii**) Finding unknown new points with the help of existing four points ((a, b), (a, b + 1), (a + 1,b), (a + 1,b + 1)), (**iv**) Enhanced image pixel representation. (**i**), (**ii**), (**iii**), (**iv**)

3.4 Applying HPF Unsharp mask filter and SRGAN for image sharpening and reconstruction

The standard SRGAN architectures provides the super-resolution module. The generator network incorporates conditional input information to better guide the enhancement process. CGAN refers the adapts model due to the integration of standard SRGAN rather than an entirely new architecture. SRGAN is an adversarial training-based deep learning model mainly used for image super-resolution [25, 26]. The SRGAN comprises generator and discriminator networks as shown in Fig. 4. The SRGAN helps to preserve the crucial details (texture and structure) and overcomes the vanishing gradient problem. The difference between the low-resolution input and high-resolution output can be learned using the residual connections. The discriminator acts as a classifier that classifies the actual (normal input) and the super-resolution image. The generator and discriminator compete with each other via multiple iterations. When the SRGAN reaches the trade-off, then the generator yields a super-resolution image as output. The adversarial training and residual learning help the SRGAN to generate a high-resolution image.

The generator architecture is similar to the Dense skip connection-based neural network (SRDenseNet) [27] and it is partitioned into 4 layers. The four layers in the generator network are the low-level feature extraction layer, the high-level feature extraction layer, the deconvolution layer, and the reconstruction layer. To improve the resolution of the image, the deconvolution layer uses two trained subpixel convolutional layers. The main generator component is the high-level feature extraction layer. The discriminator structure of the SRGAN model is similar to the CNN and it uses the Leaky Rectified linear unit (ReLU) as the activation function to prevent the negative output with a total of 8 convolutional layers. After the second convolutional layer, each layer has a batch normalization layer attached to it. This step helps to improve the network stability and processing speed.

The image reconstruction is usually taken as an optimization problem for the noisy image. This issue mainly arises due to the different types of artifacts present in the image. Let us assume that the image noise is additive which Can be addressed using a linear convolution operation without an unknown kernel. A nonlinear gamma correction can be applied to solve the pixel saturation and gamma correction problems. The corrupted noisy image can be approximated as follows:

Table 3 (i) Input image matrix (ii) BIL enhanced matrix (iii) WBI enhanced matrix

12	7	2	5	8
13	8	3	6	9
14	9	4	7	10
11	9	8	8	8
8	10	12	9	6

20	10	8	14	14
22	12	10	16	16
22	15	14	17	17
19	20	19	16	16
19	20	19	16	16

12	2	8
14	4	10
8	12	6

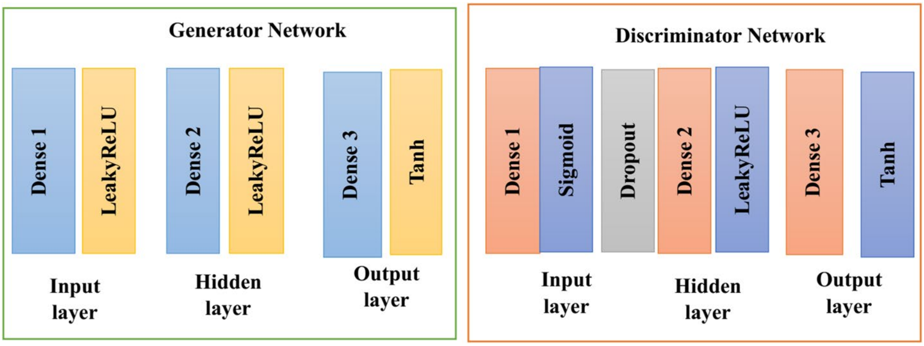

Fig. 4 SRGAN architecture

$$C\left(x\right)=F\left[\left(z+f(x)+a\right)^{\alpha}\right] \tag{5}$$

Where F is a generalized nonlinear function, a is the additive noise, z is the approximation value of the induced motion blur, and α value captures the over and under-exposed regions. The SRGAN is used for realistic image restoration. The saturation or low contrast issues mainly rise due to the higher distance between the imaged tissue and the light source. The low contrast or saturated image pixels can be noted in large areas of the image. To minimize the saturation in an image, the SRGAN uses $l_2$contextual loss. The use of natural IoT images in the training set raises the need for a correct coloration shift and this is done by applying color transfer via a linear transformation as shown as follows:

$$R`_{target}=\sum{}_{src}^{1/2}\sum{}_{target}^{-1/2}\left(R_{target}-M_{target}\right)+M_{source} \tag{6}$$

Where the covariance of the source image, the covariance of the target image, the reconstructed output, the mean of the target image, the mean of the output image, and the source image are represented as $\sum{}_{source}$, $\sum{}_{target}$, $R`_{target}$, M_{target}, and M_{source}. We have computed the covariance, mean, and source values using image intensities less than 90% of the maximum intensity value to prevent the color retransfer. The specularity, illumination, and other misc cause strong spots in the image due to the reflection from the shiny organ surfaces and bubbles. A multicolor chromatic effect is created with waterlike substances and this occurs due to the integration of linear and non-linear noise. The inconsistencies in the image appear as a missing pixel problem. The loss function designed for this problem prevents over-smoothing during mixed pixel generation.

Image readability with better results can be achieved from high-frequency components. An unsharp mask filter is a method used for enhancement of the sharpness of the image which in turn uses a high pass filter. HPF was found to be sensitive to noise, and endoscopy images due to its structural process and capture of images from disturbed surroundings leading to the production of noise. To eliminate noise from the unsharp mask filter, the noise from HPF was filtered by subtracting the normalized intensity value from the low passed determined intensity value. The mathematical expression is given below,

$$P_S(x,y)=\left(I_i(a,b)-LPF\left(I_i(a,b)\right)\right)\times\ C_n\left(5\right) \tag{7}$$

where $I_i(a,b)$ is the initial image's normalized intensity value, LPF($I_i\,(a,b)$) is the low pass filter's intensity component value and C_n is the constant amount which will be taken as the threshold value, C_n = 0.7 to 0.95 is the best-fit value after several trial and error methods. The low pass filter here used is the Gaussian filter. The inverse of image (V) was taken and the output was given to the color restoration process. Following the IE process, the RGB color space value of the image was retrieved by converting the image from HSV to RGB.

To improve the sharpness of the image, the HPF sharpness loss is integrated with the Conditional Generative Adversarial Network (CGAN) and content loss. The integrated loss function is derived as follows:

$$L_{overall} = \omega\,_1 L_{HPF} + \omega\,_2 L_{adv} + \omega\,_3 L_{content} \tag{8}$$

Where the HPF, adversarial, and content loss is represented as L_{HPF}, L_{adv}, and $L_{content}$. The weights to balance the weight of different losses are represented as ω . The L_{HPF} is also known as the sharpness loss measures the deviation between the high pass filtered image and the intensity of the actual image to improve the edges. The content loss measures whether the generated image contains the crucial details present in the actual image. The adversarial loss measures whether the discriminator can correctly distinguish the actual image from the generated image.

3.5 Cloud IoT implementation of the proposed model

The cloud setup is mainly done using Google Cloud and a virtual machine is created to run the proposed application on an OS. The edge computing capabilities are provided using the Raspberry Pi and endoscopy devices. For cloud data storage, Google Cloud is used to update the images acquired from both the dataset and endoscopy devices. The additional information is stored via the database. The proposed model is trained using the Google AI platform and hosted using the Google Kubernetes engine. Initially, the image is uploaded to the cloud storage and an azure function is executed for image processing. To minimize the latency, the initial preprocessing tasks are mainly performed in the edge device using the lightweight version of the model.

The cloud IoT integration takes place via Google Cloud IoT core service. This helps to send the processed data to the cloud for further processing. Next, the RGB to HSV transform is performed and the CLAHE with bilinear interpolation is applied. The image is sharpened using an unsharp masking filter and SRGAN model for noise suppression. Next, the image is flipped converted from HSV to RGB, and saved in the cloud. After the image is sharpened, then a notification is sent to them to download the image. The output is saved locally in the cloud and the metadata is updated in the cloud database. The performance of the proposed model is monitored using Google Cloud monitoring and regular updates are received for model improvement and firmware update. The doctors can track the patient's health history via the updated image using the IoT devices. Figure 5 presents the implementation of the proposed model in the cloud. The summary of the proposed algorithm is given below:

1 st step: Read the input image I_{mas} of size 520 × 520

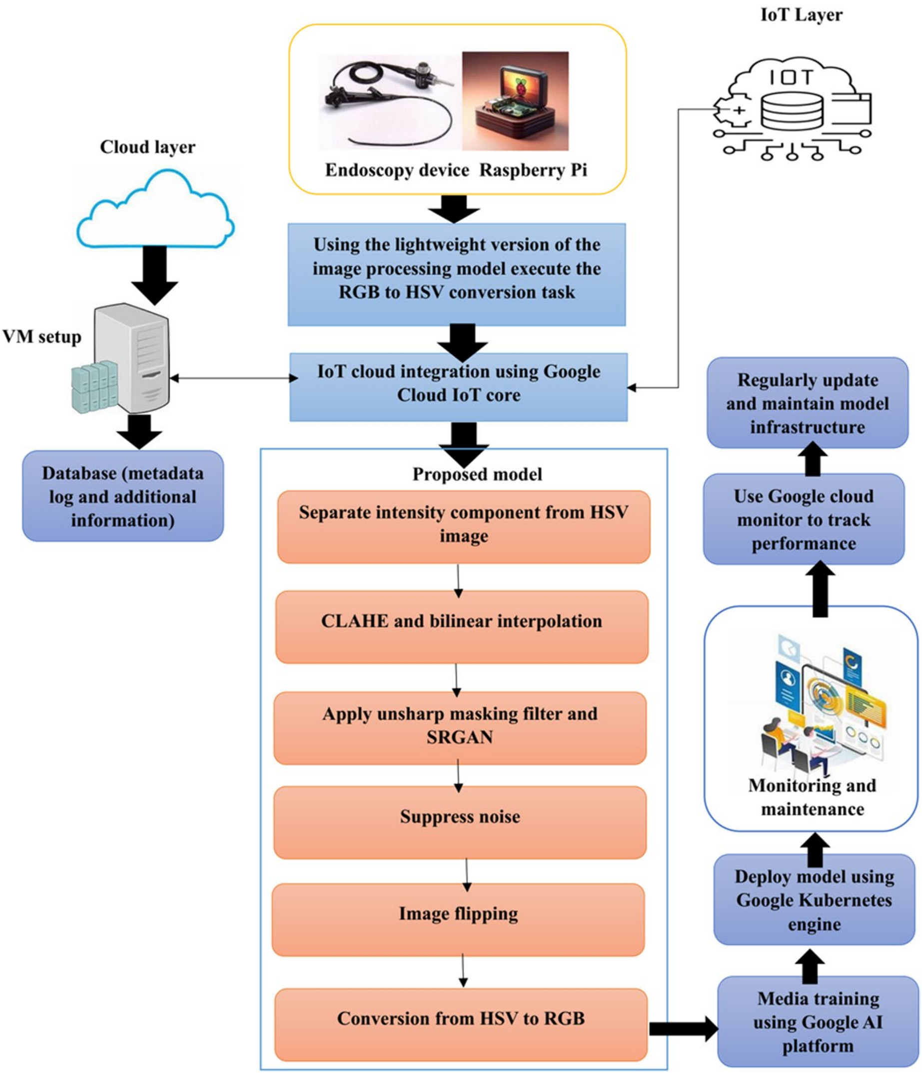

Fig. 5 Cloud IoT implementation of the proposed model

Resize image to 720 × 720 to satisfy network input constrainst.
2nd step: Conversion of the Input image from RGB to HSV color space.
3rd step: Separate the Value component from HSV and calculate the normalized Value component and perform the inverse of it.
4th step: Apply CLAHE(Contrast Limited Adaptive Histogram Equalization) with Hu-WBI to the image obtained from step 3.
By figuring out the calculated mean distance among the specified values for pixels, one can compute bilinear interpolation.
New pixel = 1/nth (Ai) + 1/nth (Bi), nth value is the distance unit between two known pixels to new unknown pixels.

5th step: Apply Unsharp Masking Filter which is a HPF to the image obtained from step 4.
6th step: Subtract the Low Pass filter's Value component with the HPF Value component to suppress noise from the image.
7th step: Obtain the image that is produced, then flip it.
8th step: Convert the obtained image from HSV to RGB and output the image.

3.6 Vision assessment

Vision assessment of endoscopic images is a challenging task since there are no golden standards or ground truth samples widely available for implementation. The surgical field visual quality is mainly addressed by the surgeons by manually verifying it. The quantitative objective assessment methods also help the researchers to evaluate image quality. Hence, in this paper, we use four metrics for visual assessment such as contrast, sharpness, naturalness, and an integration of these. The structural information (contours and boundaries) present in the endoscopic images can be accessed using the sharpness metric (δ). For δ, a maximum local variation distribution's standard deviation value is obtained.

The naturalness metric (α) identifies how the endoscopic image is realistic and it is usually a subjective metric. The naturalness value computation is often hard and it can be computed easily by taking a statistical analysis of 1000 images. The natural images usually have Gaussian and beta probability distributions. The contrast value (β) mainly identifies the variance illumination which makes the crucial details in the image easily distinguishable. The contrast is the deviation between the pixel I(u, v) and the average value of the edge weights η(u, v) in the region R.

$$\beta = P^{-1}\sum{}_{R(u,v)}\frac{|\mathrm{I}\left(u,v\right)-\eta\left(u,v\right)|}{|\mathrm{I}\left(u,v\right)+\eta\left(u,v\right)|} \tag{9}$$

$$\eta\ \left(u,v\right)=\sum{}_{R(u,v)}\mathrm{d}\left(\mathrm{u},\mathrm{v}\right)\mathrm{I}\left(u,v\right) \tag{10}$$

Where the detected edge and pixel number are represented as $\mathrm{d}\left(\mathrm{u},\mathrm{v}\right)$ and P. The hybrid metric which integrates the four visual metrics is derived as follows:

$$H=b_1\delta^{\,s_1}+b_2\alpha^{\,s_2}+\left(1-b_1-b_2\right)log\beta \tag{11}$$

Where b_1 and b_2 achieves a trade-off between the sharpness, naturalness, and contrast components. S_1 and s_2 are the parameters that control their sensitivities.

4 Experimental results and discussions

The experimental findings used to illustrate the suggested algorithm's functionality are presented in this section. The proposed algorithm was implemented to endoscopic images obtained from clinic and online datasets. Endoscopy and colonoscopy images in the upper and lower gastrointestinal tract of the human body were collected.

4.1 Experimental setup

Images and algorithms are executed using the Spyder IDE of the Anaconda framework. The hardware used in the experiments was an Intel i5-1035G1 CPU @ 1.00 GHz 1.19 GHz processor with 8 GB RAM. Table 4 displays the computation time as well as the metrics' average values. Table 4 represents the parameter setting values.

4.2 Performance metrics

Before evaluating performance metrics, the results enhanced by the proposed method and other image enhancement methods were visualized. The enhanced results for two different images are shown in Figs. 6 and 7. The visual results shown in Fig. 6 (i) and Fig. 6 (ii) exhibit the results of BPDFHE which is ineffective in overcoming halo effects and generated over-brightness artifacts in some regions, so the related histograms, which are located in the higher Value region, have an uneven distribution. The effectiveness of the suggested method to improve the details of the darker sections and more effectively eliminate halo effects was demonstrated in Fig. 6 (iii) and 6 (iv). Then, the proposed method was evaluated using standard metrics that reflected image quality. Common metrics for measuring picture quality include the feature similarity index (FSIM) and the structurally related similarity index (SSIM) which were used along with Colour enhancement factor (CEF), Value Restricted Average Local Entropy (IRMLE), and Lightness-Order-Error (LOE) used for evaluation. Here CEF indicated the measurement of quality in terms of colour enhancement. The better the color quality of an image, the higher the values of CEF.LOE was used to measure the naturalness preservation of enhanced image.

4.3 Comparative analysis

The values of metrics IRMLE, MSE, SSIM, FSIM, CEF, and LOE for 110 test images are displayed in Table 5 (one sample image). The CLAHE method has high-quality IRMLE values but its CEF and LOE values as lower. ADJ has a higher IRMLE value but its PSNR and SSIM score are lower. BPDFHE shows improved performance in terms of SSIM and LOE but its IRMLE value was lower, hence this technique could not be used for getting satisfactory performance and illumination enhancement. AHE had a better value of SSIM but its IRMLE and CEF values were lower, and not suitable for endoscopy images as shown in Table 4; Fig. 6. The proposed algorithm performs exceptionally well compared to the state-of-the-art techniques. The proposed algorithm offer optimal results in eliminating and reducing noise than the other methods and its suitability for real-time applications is high.

To guarantee quality improvement of endoscopy images and to view the detailed structures and vessels without altering the similarity to the original reference images, differ-

Table 4 Parameter setting

Parameters	Values
Clip limit	2
Learning rate	1e-4
Residual blocks	16
Tile size	8 × 8
Loss weights	0.1
Gamma Corrections	0.8

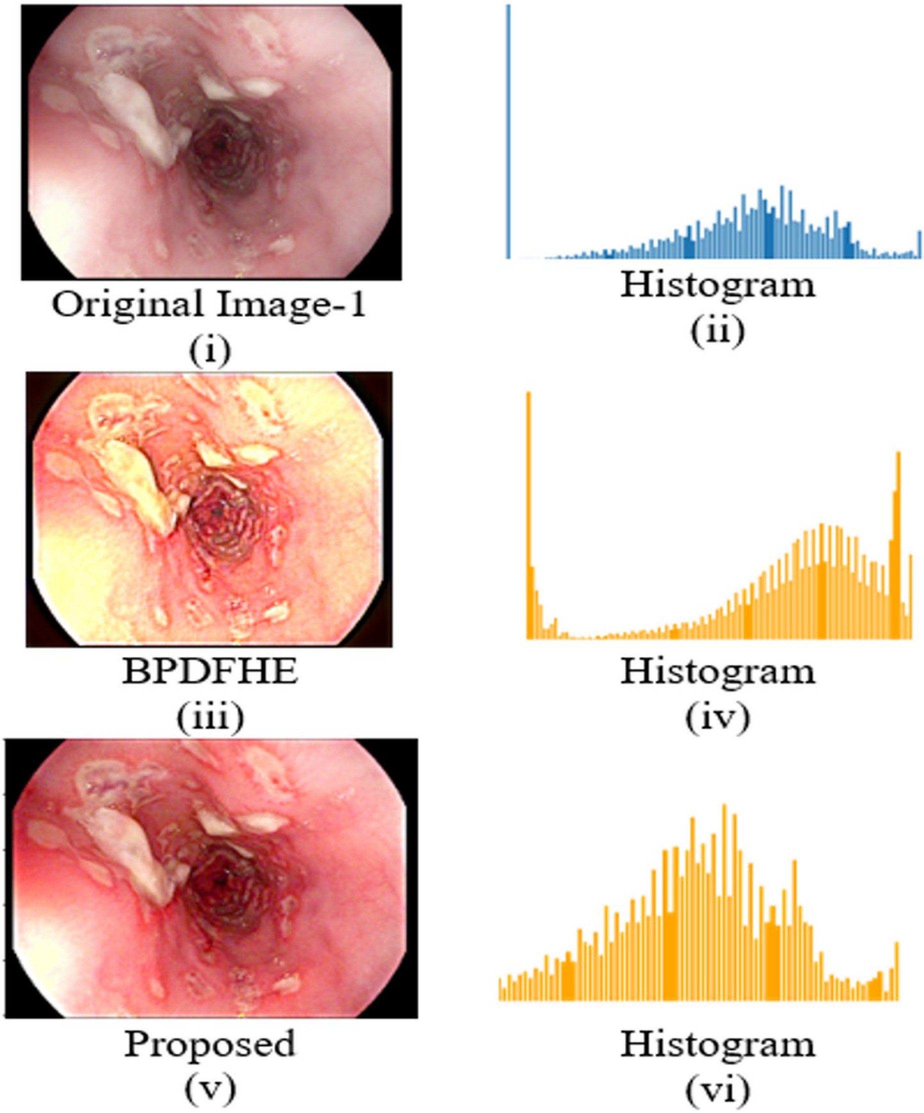

Fig. 6 Findings from comparisons between the suggested method and other cutting-edge techniques (**i**) original Image-1 and (**ii**) histogram of original image-1 (**iii**) BPDFHE enhanced and (**iv**) its histogram (**v**) enhanced by proposed method and (**vi**) its histogram

ent metrics such as Structural Similarity Index (SSIM), Mean Squared Error (MSE), and Adjusting image Intensity values (ADJ) was utilized to compare with different methods such as Histogram Equalisation (HE), AHE, and Adjusting image Intensity values (ADJ). The results are shown in the following figures. The comparison of IRMLE values or various image enhancement methods are as depicted in Fig. 8.The superior reconstrcuction accuracy is indicated and the proposed model accomplishes highest IRMLE value contrasted with existing approaches as shown in Fig. 8.

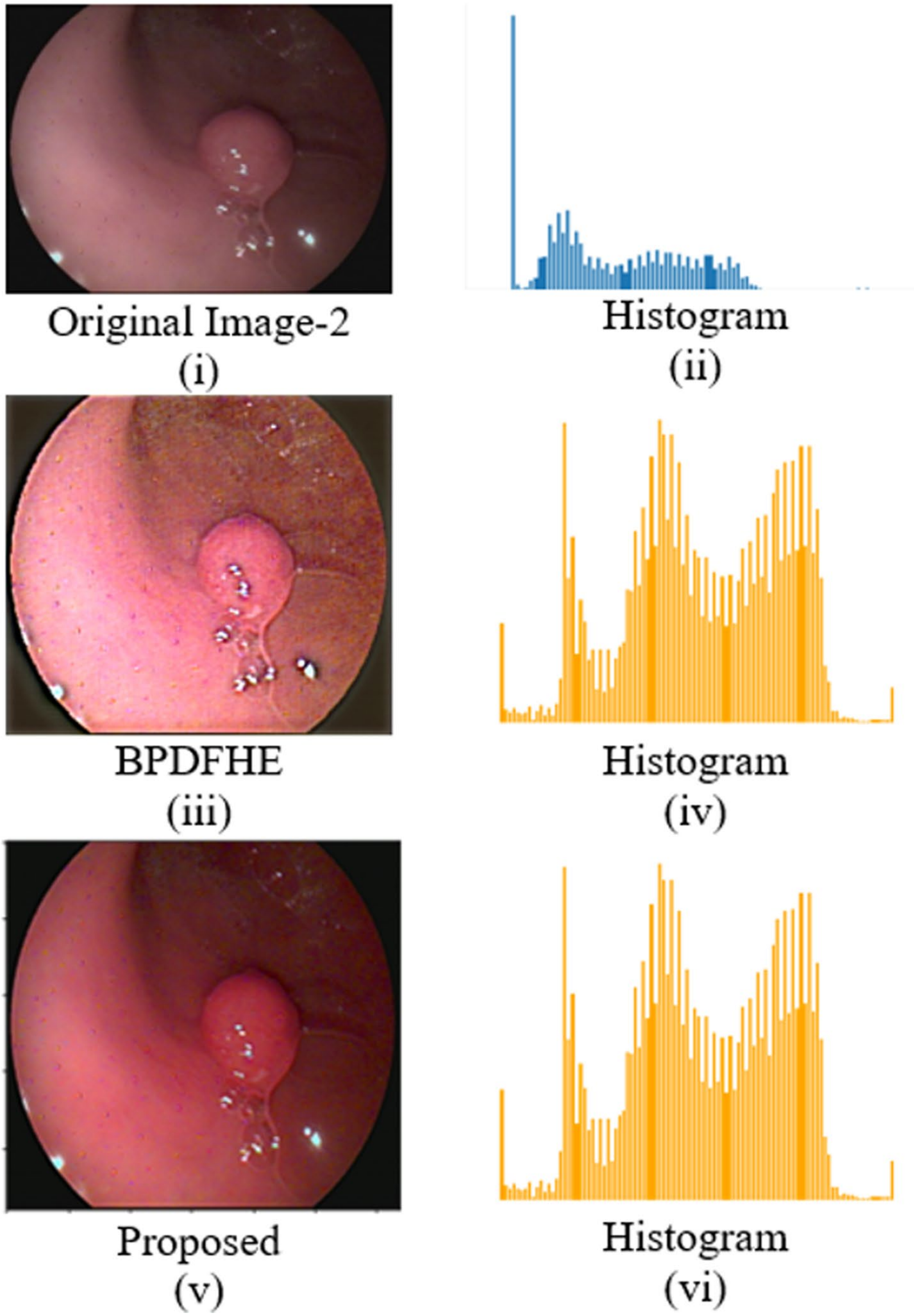

Fig. 7 Results of comparisons enhanced by the proposed method with other state-of-the-art methods (**i**) original Image-1 and (**ii**) its histogram. (**iii**) BPDFHE enhanced with (**iv**) its histogram (**v**) enhanced by the proposed method and (**vi**) its histogram

Table 5 Evaluation metrics for a proposed method with other image enhancement algorithms

Methods tested	Original	CLAHE	BPDFHE	AHE	ADJ	Proposed
IRMLE	1.0	1.35	0.92	0.92	0.98	1.32
SSIM	1.0	0.74	0.82	0.72	0.75	0.86
FSIM	-	0.724	0.802	0.762	0.831	0.912
CEF	1	1.15	1	0.17	0.014	1.12
LOE	0	1.54	1.75	12.26	1.86	1.03
PSNR	Inf.	20.09	23.16	16.47	26.14	27.32
Computational time (s)	0.01	0.18	0.35	0.22	0.15	1.12

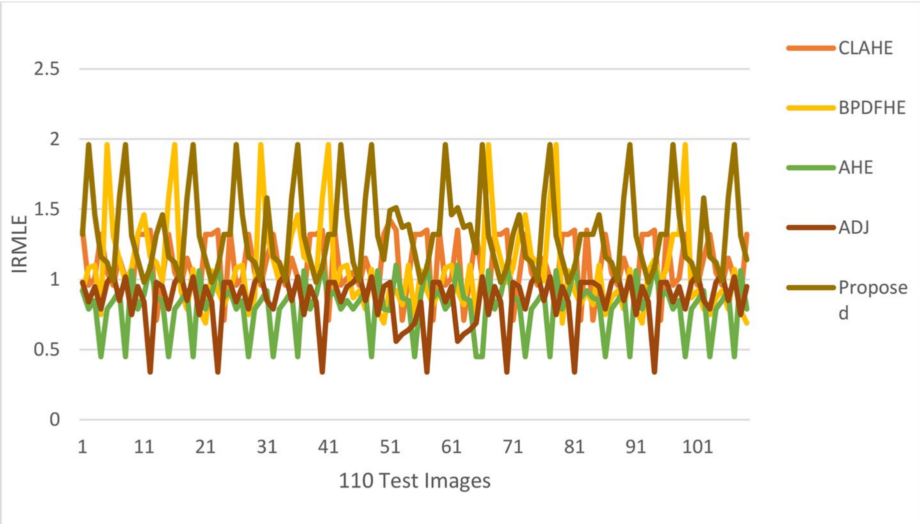

Fig. 8 Comparison of IRMLE values or various image enhancement methods

The IRME values of the proposed model are compared with existing techniques such as CLAHE, BPDFHE, AHE, and ADI. The IRMLE (Image Reconstruction Mean Logarithmic Error) mainly evaluates the accuracy and quality of the reconstructed image. The results show that the proposed model offers higher IRME values due to the incorporation of the CLAHE with bilinear interpolation, CGAN, and an unsharp masking filter.

The LOE (Logarithmic Over-Enhancement) values of the proposed model are compared with the existing techniques and the results are demonstrated in Fig. 9. A minimal LOE value indicates that the model does not offer over-enhancement and based on the results we can conclude that the proposed model does not introduce additional noise or artifacts due to the different methodologies utilized. The ADI and AHE techniques offer higher LOE value which shows their ability to preserve the naturality of the image is low.

The SSIM of the proposed model is compared with the existing techniques and the results are shown in Fig. 10. SSIM value which is very close or equal to 1 can be considered as a higher value which in turn, presents the improved quality by comparing the similarity of the enhanced and reference images. The proposed model offers a higher SSIM value showing

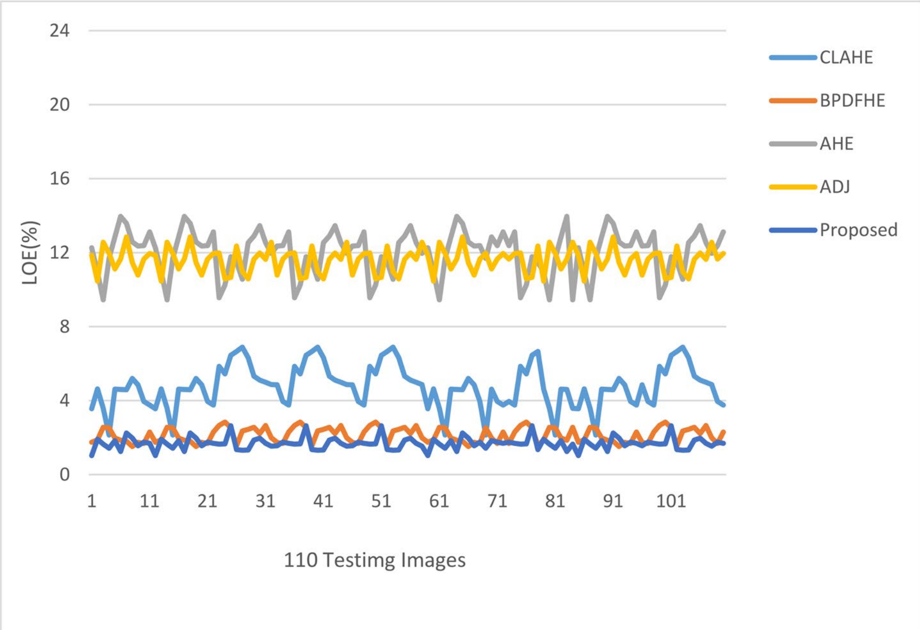

Fig. 9 LOE comparison of various image enhancement methods

Fig. 10 SSIM comparison of various image enhancement methods like CLAHE, AHE, ADJ, BPDFHE and proposed work

Table 6 Quantitative comparisonof the proposed model with existing methods over CVC clinic and real-time dataset

Technique	PSNR (Db)	SSIM	LPIPS
EEO [12]	14.3255	0.6564	0.2312
GFD-WT [13]	19.5235	0.7645	0.2982
Specular reflection removal [14]	18.3432	0.6123	0.3123
HE [15]	17.4550	0.5142	0.2212
WGF [17]	19.5234	0.5621	0.1942
Retinex [18]	21.3555	0.6110	0.1723
Proposed	26.4365	0.9232	0.1324

Table 7 Comparative analysis in real-time dataset

Technique	PSNR (Db)	SSIM	LPIPS
EEO [12]	21.4365	0.7751	0.2012
GFD-WT [13]	19.4365	0.7032	0.1990
Specular reflection removal [14]	13.2412	0.6541	0.3012
HE [15]	15.5523	0.6651	0.2312
WGF [17]	21.4343	0.7012	0.2012
Retinex [18]	23.5644	0.7211	0.1901
Proposed	26.4365	0.9552	0.1308

that it preserves the image structure and image quality accurately. As per the results, using the CLAHE technique alone does not retain the structural integrity of the actual image.

The proposed model is compared with different existing techniques such as EEO [12], GFD-WT [13], Specular reflection removal [14], HE [15], WGF [17], and Retinex [18] using the CVC clinic and real-time dataset and the results are presented in Tables 6 and 7. The comparisons are made using PSNR, SSIM, and LPIPS (Learned Perceptual Image Patch Similarity). The perceptual similarity between images is measured using LPIPS and it is a valid measure to evaluate the model's efficiency to capture the perceptual differences. The proposed model offers minimal LPIPS values on both datasets which shows that the two images are perceptually similar to each other. This shows that the images are identical in terms of human perception.

The proposed model offers a PSNR value of 26.4365 and 23.4365 on the CVC clinic and real-time dataset. The SSIM values of the proposed model on the CVC clinic and real-time dataset are 0.9232 and 0.9552 respectively. The results show that the proposed model offers improved performance compared to the existing algorithms. The retinex model offers the next best performance when compared to the remaining existing techniques. The results demonstrate that the proposed model minimizes the disparities and offers superior structural resemblance due to the CGAN and HPF techniques incorporated. The proposed model improves the luminance in the central recessed region and maintains the color consistency to improve the ability of the physician to analyze complex physiological structures.

4.4 Ablation study

To evaluate the efficiency of each component used in the proposed methodology we have designed an ablation study. Initially, the ablation experiment is conducted by leaving certain modules and then leaving the CGAN losses. The experiments for the ablation study are conducted on the CVC clinic database. The different performance evaluation metrics used are PSNR, SSIM, and LPIPS. Table 8 presents the ablation study conducted by uti-

Table 8 Ablation experiments conducted analyzing different components during training

Technique	PSNR (Db)	SSIM	LPIPS
RGB to HSV conversion (C1) + HPF	19.3141	0.8765	0.1501
C1 + CLAHE (C2) + HPF	20.1212	0.8945	0.1456
C1 + CLAHE- Hu-WBI (C3) + HPF	22.1231	0.9012	0.1415
C1 + C3+HPF unsharp mask filtering (C4)	23.1212	0.9123	0.1390
Proposed	26.4365	0.9232	0.1324

Table 9 Ablation study using adversarial and content loss functions

Sequence	L_{HPF}	$L_{adv}+L_{content}$	PSNR (Db)	SSIM
1	✓		24.2134	0.9345
2	✓	✓	26.4365	0.9232

Table 10 Statistical evaluation of proposed model

Technique		EEO [12]	GFD-WT [13]	Retinex [18]	Proposed
PSNR	**Mean ± SD**	20.09 ± 0.92	16.47 ± 1.12	26.14 ± 0.79	26.43 ± 0.91
	p-value	< 0.001	0.002	< 0.001	-
SSIM	**Mean ± SD**	0.74 ± 0.021	0.75 ± 0.025	0.61 ± 0.023	0.923 ± 0.015
	p-value	< 0.001	0.003	< 0.001	-
LPIPS	**Mean ± SD**	0.221 ± 0.010	0.165 ± 0.008	0.240 ± 0.0012	0.132 ± 0.009
	p-value	< 0.001	0.0014	< 0.001	-

lizing different components. The components C1, C2, C3, and C4 represent the RGB to HSV conversion, CLAHE, CLAHE + Hu-WBI algorithm, and HPF unsharp mask filtering. The HPF is used in every step to create a super-resolution image. The C1 + CLAHE (C2) + HPF method offers a PSNR, SSIM, and LPIPS value of 20.1212, 0.8945, and 0.1456. The C1 + CLAHE- Hu-WBI algorithm (C3) + HPF combination offers a PSNR, SSIM, and LPIPS value of 20.1212, 0.8945, and 0.1456. The results show that the improvement of the proposed model is mainly due to the incorporation of the CGAN model. When C1, C2, and C3 were used individually it resulted in higher LPIPS values and lower SSIM and PSNR values. In Table 9, improved PSNR and SSIM values are only obtained when the adversarial and content loss function is included along with the HPF loss function. The ablation study conducted shows the efficiency of each module and loss function in improving the low-light endoscopic images. The efficiency of the proposed model can be identified in terms of noise processing, preserving the color fidelity, brightness, and sharpness details.

Table 10 explains the statistical evlaluation of proposed model along with existing methods like EEO [12], GFD-WT [13], Retinex [18]. The mean± standard deviation value and p-values are computed in this statistical evlautaion. Compared to the previous techniques, the proposed work offers better results.

5 Conclusion

In this study, we suggested an image-enhancing technique for endoscopic pictures with different contrast. To address low contrast, overexposure light effect, and darker portions of the image, the suggested solution applies CLAHE and Hu-WBI to the inverse of the intensity image component. The outcomes of many trials, such as SSIM, IRMLE, LOE, and FSIM showed the proposed method enhancing illumination, details of the darker areas of the images, and reducing the effect of over-exposure brightness of light. In addition, the proposed method used an unsharp masking filter for sharpening the image along with an effective noise reduction. When compared to current techniques, the suggested method's efficiency in general is good and satisfactory. This work presents the four-stage endoscopic image enhancement image. For contrast enhancement using CLAHE, process input RGB images are converted Value (V) channel and HSV color space. An unsharp masking and Hu-WBI performs edge refinement and artifacts suppression. Finally, apply SRGAN based super-resolution to reconstruct the enhanced image, improving perceptual quality and preserving structural details. The conventional techniques are outperformed using proposed model demonstrates visual evaluations and extensive quantitative in terms of visual clarity, SSIM and PSNR. The efficiency of the proposed model is validated in real-time by deploying it in an IoT-based cloud environment. In the future, we plan to adjust the proposed model to be applied to multimodal image modalities.

Authors' contributions **Author 1**: Participated in the development of the research methodology, including determining the study design, data analysis methods, and tools used. **Author 2**: Contributed to the overall research concept, including designing the framework and shaping the scope of the study.

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Data availability Authors are working on implementing the same using real world data with appropriate permissions.

Declarations

Ethics approval and consent to participate Not Applicable.

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Competing interests The authors declare that we have no competing interest.

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